Faculty of Computer Science and Mathematics
Universiti Malaysia Terengganu

LAB REPORT

CSE3023 – Web-Based Application Development

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Semester	4
Lab Report	2
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2.9 Exercise

Objective: Develop a Product Inventory System by applying knowledge of Servlets, JavaBeans, and the DAO pattern to create a simple dynamic web application.

Instruction

Please complete this exercise by fulfilling the following requirements:

1. Database Setup: Create a new table named products in your CSE3023 database with the following columns:
    - id (INT, Primary Key, Auto Increment)
    - name (VARCHAR)
    - category (VARCHAR)
    - price (DOUBLE)
    - quantity (INT)

QUERY

`USE cse3023;`
 	
`CREATE TABLE products (`
`  id INT AUTO_INCREMENT PRIMARY KEY,`
`  name VARCHAR(100),`
`  category VARCHAR(100),`
`  price DOUBLE,`
`  quantity INT`
`);`

2. JavaBean (Model): Create a class named Product in the com.lab.model package. Ensure it has the appropriate private attributes, constructors (empty and parameterized), and getter/setter methods.

QUERY
Product.java

```java
package com.lab.model;


/**
 * Name: Nurbatrisyia Aliza binti Alias
 * Program: SMSK(SE)
 * Course: CSE3023
 * Lab: MP1
 * Date: 07/04/2026
 */


public class Product {
    private int id;
    private String name;
    private String category;
    private double price;
    private int quantity;

    // Empty constructor
    public Product() {}

    // Constructor without ID (Insert)
    public Product(String name, String category, double price, int
quantity) {
        this.name = name;
        this.category = category;
        this.price = price;
        this.quantity = quantity;
    }
```

```
    // Constructor with ID (Update)
    public Product(int id, String name, String category, double
price, int quantity) {
        this.id = id;
        this.name = name;
        this.category = category;
        this.price = price;
        this.quantity = quantity;
    }

    // Getter & Setter
    public int getId() { return id; }
    public void setId(int id) { this.id = id; }

    public String getName() { return name; }
    public void setName(String name) { this.name = name; }

    public String getCategory() { return category; }
    public void setCategory(String category) { this.category =
category; }

    public double getPrice() { return price; }
    public void setPrice(double price) { this.price = price; }

    public int getQuantity() { return quantity; }
    public void setQuantity(int quantity) { this.quantity =
quantity; }
}
```

3. Data Access Object (DAO): Create a ProductDAO class in the com.lab.dao package to handle all SQL operations (INSERT, SELECT, UPDATE, DELETE).

QUERY

ProductDAO.java

```
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
 * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this template
 */
package com.lab.dao;

/**
 * Name: Nurbatrisyia Aliza binti Alias
 * Program: SMSK(SE)
 * Course: CSE3023
 * Lab: MP1
 * Date: 07/04/2026
 */

import com.lab.model.Product;
import java.sql.*;
import java.util.ArrayList;
import java.util.List;

public class ProductDAO {

    private String jdbcURL = "jdbc:mysql://localhost:3306/CSE3023";
    private String jdbcUsername = "root";
    private String jdbcPassword = "admin";

    protected Connection getConnection() {
```

```
        Connection conn = null;
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            conn = DriverManager.getConnection(jdbcURL,
jdbcUsername, jdbcPassword);
        } catch (Exception e) {
            e.printStackTrace();
        }
        return conn;
    }

    // CREATE
    public void insertProduct(Product product) {
        String sql = "INSERT INTO products (name, category, price,
quantity) VALUES (?, ?, ?, ?)";
        try (Connection conn = getConnection();
             PreparedStatement ps = conn.prepareStatement(sql)) {

            ps.setString(1, product.getName());
            ps.setString(2, product.getCategory());
            ps.setDouble(3, product.getPrice());
            ps.setInt(4, product.getQuantity());

            ps.executeUpdate();
        } catch (Exception e) {
            e.printStackTrace();
        }
    }

    // READ ALL
    public List<Product> selectAllProducts() {
        List<Product> list = new ArrayList<>();
        String sql = "SELECT * FROM products";
```

```
    try (Connection conn = getConnection();
         PreparedStatement ps = conn.prepareStatement(sql)) {

        ResultSet rs = ps.executeQuery();

        while (rs.next()) {
            list.add(new Product(
                    rs.getInt("id"),
                    rs.getString("name"),
                    rs.getString("category"),
                    rs.getDouble("price"),
                    rs.getInt("quantity")
            ));
        }
    } catch (Exception e) {
        e.printStackTrace();
    }
    return list;
}

// READ ONE
public Product selectProduct(int id) {
    Product p = null;
    String sql = "SELECT * FROM products WHERE id=?";

    try (Connection conn = getConnection();
         PreparedStatement ps = conn.prepareStatement(sql)) {

        ps.setInt(1, id);
        ResultSet rs = ps.executeQuery();

        if (rs.next()) {
```

```
                p = new Product(
                        id,
                        rs.getString("name"),
                        rs.getString("category"),
                        rs.getDouble("price"),
                        rs.getInt("quantity")
                );
            }
        } catch (Exception e) {
            e.printStackTrace();
        }
        return p;
    }

    // UPDATE
    public boolean updateProduct(Product p) {
        String sql = "UPDATE products SET name=?, category=?,
price=?, quantity=? WHERE id=?";
        boolean updated = false;

        try (Connection conn = getConnection();
             PreparedStatement ps = conn.prepareStatement(sql)) {

            ps.setString(1, p.getName());
            ps.setString(2, p.getCategory());
            ps.setDouble(3, p.getPrice());
            ps.setInt(4, p.getQuantity());
            ps.setInt(5, p.getId());

            updated = ps.executeUpdate() > 0;
        } catch (Exception e) {
            e.printStackTrace();
        }
```

```
        return updated;
    }

    // DELETE
    public boolean deleteProduct(int id) {
        String sql = "DELETE FROM products WHERE id=?";
        boolean deleted = false;

        try (Connection conn = getConnection();
             PreparedStatement ps = conn.prepareStatement(sql)) {

            ps.setInt(1, id);
            deleted = ps.executeUpdate() > 0;
        } catch (Exception e) {
            e.printStackTrace();
        }
        return deleted;
    }
}
```

4. Servlets (Controller): Create the necessary Servlets (e.g., InsertProductServlet, ViewProductServlet, UpdateProductServlet, DeleteProductServlet) to capture user inputs and interact with your ProductDAO. Strict rule: No JDBC code should be written inside these Servlets.

QUERY
InsertProductServlet.java

```
/*
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 * Click nbfs://nbhost/SystemFileSystem/Templates/JSP_Servlet/Servlet.java to edit this template
 */

package com.lab.controller;

import com.lab.dao.ProductDAO;
import com.lab.model.Product;
import java.io.IOException;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

/**
 * Name: Nurbatrisyia Aliza binti Alias
 * Program: SMSK(SE)
 * Course: CSE3023
 * Lab: MP1
 * Date: 07/04/2026
 */
public class InsertProductServlet extends HttpServlet {
```

```
    private ProductDAO dao;

    public void init() {
        dao = new ProductDAO();
    }

    protected void doPost(HttpServletRequest req,
HttpServletResponse res)
            throws ServletException, IOException {

        String name = req.getParameter("name");
        String category = req.getParameter("category");
        double price =
Double.parseDouble(req.getParameter("price"));
        int quantity =
Integer.parseInt(req.getParameter("quantity"));

        Product p = new Product(name, category, price, quantity);
        dao.insertProduct(p);

        res.sendRedirect("ViewProductServlet");
    }
}
```

ViewProductServlet.java

```
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-
default.txt to change this license
 * Click
nbfs://nbhost/SystemFileSystem/Templates/JSP_Servlet/Servlet.java
to edit this template
 */

package com.lab.controller;
```

```
import com.lab.dao.ProductDAO;
import com.lab.model.Product;
import java.io.IOException;
import java.io.PrintWriter;
import java.util.List;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

/**
 * Name: Nurbatrisyia Aliza binti Alias
 * Program: SMSK(SE)
 * Course: CSE3023
 * Lab: MP1
 * Date: 07/04/2026
 */
public class ViewProductServlet extends HttpServlet {

    private ProductDAO dao;

    @Override
    public void init() {
        dao = new ProductDAO();
    }

    @Override
    protected void doGet(HttpServletRequest req,
HttpServletResponse res)
            throws IOException {

        res.setContentType("text/html");
        PrintWriter out = res.getWriter();
```

```
        List<Product> list = dao.selectAllProducts();

        out.println("<html>");
        out.println("<head>");
        out.println("<title>Product List</title>");
        out.println("</head>");
        out.println("<body>");

        out.println("<h2>Product List</h2>");
        out.println("<table border='1'>");

out.println("<tr><th>ID</th><th>Name</th><th>Category</th><th>Price
</th><th>Qty</th><th>Action</th></tr>");

        for (Product p : list) {
            out.println("<tr>");
            out.println("<td>" + p.getId() + "</td>");
            out.println("<td>" + p.getName() + "</td>");
            out.println("<td>" + p.getCategory() + "</td>");
            out.println("<td>" + p.getPrice() + "</td>");
            out.println("<td>" + p.getQuantity() + "</td>");
            out.println("<td><a href='UpdateProductServlet?id=" +
p.getId() + "'>Edit</a> | ");
            out.println("<a href='DeleteProductServlet?id=" +
p.getId() + "'>Delete</a></td>");
            out.println("</tr>");
        }

        out.println("</table>");
        out.println("<br><br>");
        out.println("<a href='add_product.html'>+ Add
Product</a>");
```

```
        out.println("</body>");
        out.println("</html>");
    }
}
```

UpdateProductServlet.java

```
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-
default.txt to change this license
 * Click
nbfs://nbhost/SystemFileSystem/Templates/JSP_Servlet/Servlet.java
to edit this template
 */

package com.lab.controller;

import com.lab.dao.ProductDAO;
import com.lab.model.Product;
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

/**
 * Name: Nurbatrisyia Aliza binti Alias
 * Program: SMSK(SE)
 * Course: CSE3023
 * Lab: MP1
 * Date: 07/04/2026
 */

public class UpdateProductServlet extends HttpServlet {
```

```
    private ProductDAO productDAO;

    @Override
    public void init() {
        productDAO = new ProductDAO();
    }

    @Override
    protected void doGet(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        int id = Integer.parseInt(request.getParameter("id"));

        // Call DAO
        Product product = productDAO.selectProduct(id);

        if (product != null) {
            out.println("<h2>Update Product</h2>");
            out.println("<form action='UpdateProductServlet'
method='POST'>");

            out.println("<input type='hidden' name='id' value='" +
product.getId() + "'>");

            out.println("Name: <input type='text' name='name'
value='"
                        + product.getName() + "'
required><br><br>");
```

```
            out.println("Category: <input type='text'
name='category' value='"
                        + product.getCategory() + "'
required><br><br>");


            out.println("Price: <input type='number' step='0.01'
name='price' value='"
                        + product.getPrice() + "'
required><br><br>");


            out.println("Quantity: <input type='number'
name='quantity' value='"
                        + product.getQuantity() + "'
required><br><br>");


            out.println("<input type='submit' value='Update
Product'>");
            out.println("</form>");
        } else {
            out.println("<h3>Product not found!</h3>");
        }
    }


    @Override
    protected void doPost(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {


        int id = Integer.parseInt(request.getParameter("id"));
        String name = request.getParameter("name");
        String category = request.getParameter("category");
        double price =
Double.parseDouble(request.getParameter("price"));
        int quantity =
Integer.parseInt(request.getParameter("quantity"));
```

```
        Product product = new Product(id, name, category, price,
quantity);

        // Call DAO
        productDAO.updateProduct(product);

        response.sendRedirect("ViewProductServlet");
    }
}
```

DeleteProductServlet.java

```
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-
default.txt to change this license
 * Click
nbfs://nbhost/SystemFileSystem/Templates/JSP_Servlet/Servlet.java
to edit this template
 */

package com.lab.controller;

import com.lab.dao.ProductDAO;
import java.io.IOException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

/**
 * Name: Nurbatrisyia Aliza binti Alias
 * Program: SMSK(SE)
 * Course: CSE3023
 * Lab: MP1
```

```
 * Date: 07/04/2026
 */

public class DeleteProductServlet extends HttpServlet {

    private ProductDAO dao;

    public void init() {
        dao = new ProductDAO();
    }

    protected void doGet(HttpServletRequest req,
HttpServletResponse res)
            throws IOException {

        int id = Integer.parseInt(req.getParameter("id"));
        dao.deleteProduct(id);

        res.sendRedirect("ViewProductServlet");
    }
}
```

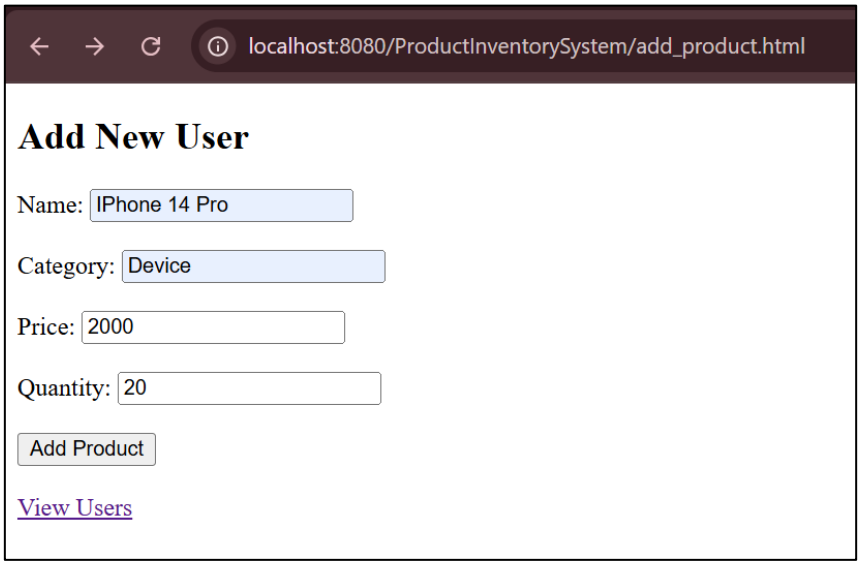
5. User Interface (View): Create an HTML form (add_product.html) to insert new products. Ensure your view servlet generates an HTML table displaying all products, along with functional ”Edit” and ”Delete” hyperlinks for each row.

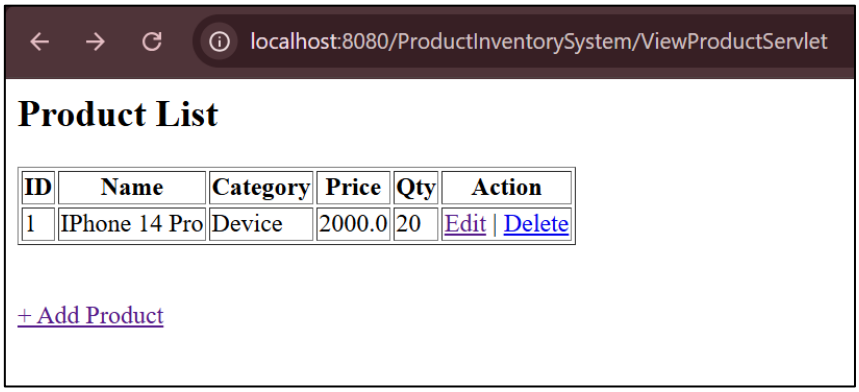
QUERY
add_product.html `<!DOCTYPE html>` `<!--` `Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license` `Click nbfs://nbhost/SystemFileSystem/Templates/JSP_Servlet/Html.html to edit this template` `-->` `<!--` `Name: Nurbatrisyia Aliza binti Alias` `Program: SMSK(SE)` `Course: CSE3023` `Lab: MP1` `Date: 07/04/2026` `-->` `<html>` `<head>` `    <title>Add User</title>` `</head>` `<body>` `    <h2>Add New User</h2>` `    <form action="InsertProductServlet" method="POST">` `    Name: <input type="text" name="name" required>  ` `    Category: <input type="text" name="category" required>  ` `    Price: <input type="number" step="0.01" name="price" required>  `

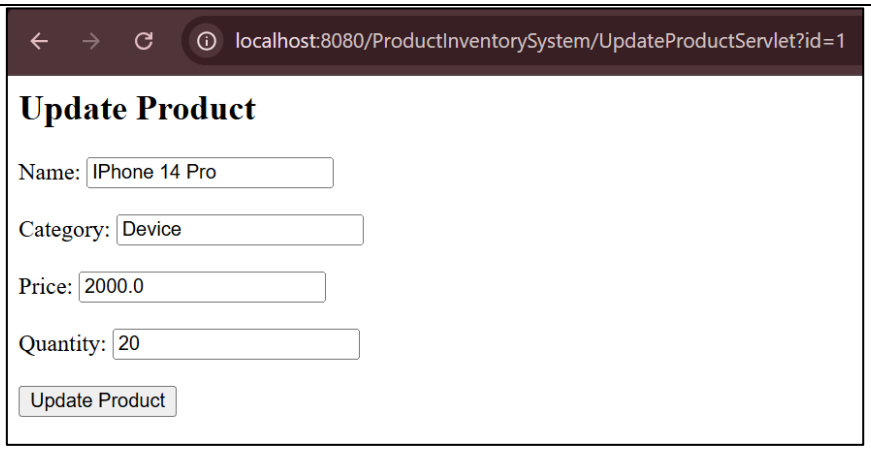
```
    Quantity: <input type="number" name="quantity"
required><br><br>
    <input type="submit" value="Add Product">
    </form>

    <br>
    <a href="ViewProductServlet">View Users</a>
</body>
</html>
```

RESULT






localhost:8080/ProductInventorySystem/UpdateProductServlet?id=1
Update Product
Name: IPhone 14 Pro
Category: Device
Price: 2000.0
Quantity: 20
Update Product